Automated Explanation-Based Trustworthiness Testing and Improvement of AI-Based Systems

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***Abstract*—AI-based systems are increasingly used in software engineering (SE). However, users still question whether their decisions are trustworthy, i.e, they are made for the right reasons. Such trustworthiness directly affects users' acceptance and systems' adoption. Despite its importance, trustworthiness testing remains underexplored. To support this, explanations have emerged as a useful basis, as reviewing them helps users understand the rationale behind the decisions. Unfortunately, manual review is not practical, given the overwhelming amount of AI-produced decisions. To address this challenge, this dissertation aims to develop automated approaches for trustworthiness testing. We first contextualize trustworthiness through the lens of explanation correctness. We then conduct empirical studies to benchmark the trustworthiness of AI-based systems across various SE tasks, such as vulnerability detection and program repair. Based on insights from the benchmark, we develop automated approaches to test and improve trustworthiness. Ultimately, this dissertation paves the way for more trustworthy and responsible AI-assisted SE.**

I. Problem and Research Statement

AI has revolutionized SE in various tasks, such as requirement engineering, issue management, vulnerability detection, and program repair. Despite their superior performance during development, AI-based systems can still fail in real-life scenarios [1]. As a result, people still raise concerns about the trustworthiness of their decisions [2].

To evaluate a system, prior studies mainly focus on its general task-solving ability using metrics, such as model confidence or accuracy. Although these metrics are useful, it is also important to understand the decision-making process [3]. Several studies have shown that these metrics alone are insufficient indicators of trustworthiness [4]. The system may rely on spurious patterns, making overconfident decisions but for the wrong reasons [5]. We call such decisions *untrustworthy*.

Untrustworthy decisions can create a false sense of reliability. Even though the outcomes look correct, the underlying reasoning is flawed. Users may be misled into believing this flawed reasoning. In addition, while the flawed reasoning spuriously leads to correct outcomes, it may not apply to real-world data. Hence, untrustworthy decisions also indicate potential failures in the real world.

In this dissertation, *trustworthiness* refers to the ability of an AI-based system to make decisions based on the correct reasons. This is an important factor that directly affects user acceptance and system adoption. However, trustworthiness testing has not received as much attention and consequently remains less resolved. The lack of trustworthiness testing approaches hinders the broader adoption of AI-based systems.

To support trustworthiness testing, model explanations have emerged as a promising basis. For example, explanations of classification systems can highlight the most important input features that steer the decisions [6]. For large language models (LLMs), chain-of-thought (CoT) prompting [7] can outline a series of intermediate reasoning steps that justify the final answer. By reviewing these explanations, users can understand the system's rationale, identifying untrustworthy decisions.

Nevertheless, the current use of explanations for trustworthiness testing remains largely manual due to the lack of automated approaches. Users must review explanations, judge whether the highlighted features or reasoning steps are appropriate, and determine whether the corresponding decisions are trustworthy. This process is time-consuming, error-prone, inconsistent, and difficult to scale to the overwhelming amount of AI-generated decisions. This limitation motivates the need for automated approaches for trustworthiness testing.

To address this gap, this dissertation investigates the trustworthiness of AI-based systems in SE and how to test it. It aims to answer three main research questions (RQs):

RQ1 How trustworthy are AI-based systems in SE?
RQ2 How can we automatically test the trustworthiness of AI-based systems in SE?
RQ3 How can trustworthiness test results be used to improve AI-based SE systems?

We first conduct human-involved empirical studies to benchmark the trustworthiness of state-of-the-art (SOTA) AI-based systems. We investigate various SE classification and generation tasks, such as text classification, vulnerability detection, and program repair. The goal is to quantify the prevalence of untrustworthy decisions, analyse their nature and patterns, and diagnose their underlying root causes. Second, we develop automated approaches for testing the trustworthiness of these systems. Third, we design automated approaches for improving trustworthiness by utilising test results.

Ultimately, this research aims to advance the development and deployment of trustworthy AI-based systems in SE. It can reduce the reliance on manual review of explanations during trustworthiness testing and provide feedback for debugging or repairing untrustworthy systems. In practice, automated

trustworthiness testing can be integrated into iterative development, continuous integration and delivery pipelines of AI-based systems. These advances will contribute to their responsible and effective adoption in SE and beyond.

II. Background and Related Work

This dissertation focuses on *trustworthiness*, in particular, whether a system makes decisions based on the correct reasons rather than spurious ones. We adopt the definition of trustworthiness proposed by Kästner et al. [8]: a system is trustworthy to a stakeholder in a given context if and only if (1) it works properly in the context and (2) the stakeholder has justified belief in it.

Prior studies mainly focus on assessing (1) by testing various aspects of an AI-based system, such as accuracy, efficiency, and fairness [9]. They evaluate the general task-solving ability, i.e., what output the system produces. However, for AI-based SE systems, (2) requires more than that. It needs to evaluate the decision-making process, i.e., why the system produces that output. Therefore, it is also important to have evidence that the decisions are based on correct reasons.

Local *explanations* (explanations for short) can provide such evidence [6]. They break down an individual decision into its components. This allows users to grasp the decision-making process in a way that aligns with human cognitive patterns. Different explainability methods have been introduced to generate explanations, depending on the systems. For classification systems, explanations commonly highlight the most important input features that steer the decisions [6]. For generative systems, such as LLMs, they can provide a series of intermediate reasoning steps before producing the final answer using CoT prompting [7]. These explainability methods have been used in many SE tasks. For example, explanations are used to provide finer-grained detection of software vulnerabilities [10]. In program repair, explanations can accelerate the repair validation process by helping users identify unreliable or hallucinatory repairs [9].

In trustworthiness testing, explanations can serve as a useful basis [8]. If an explanation highlights irrelevant features or contains hallucinations, the decision is untrustworthy. However, prior work still relies on manual explanation review. It is overwhelming for human users to review explanations manually, given the increasing amount of AI-generated decisions.

This dissertation bridges this gap by introducing automated approaches for trustworthiness testing. Crucially, it does not compete with existing explanation generation methods, but addresses a complementary problem. Instead of generating explanations, it evaluates whether the explanations generated by any method are correct and reflect meaningful reasoning. In this way, trustworthiness testing acts as an independent component that can be paired with any explainability method. Each provides a different lens on the system's behaviour, and the dissertation helps test the trustworthiness of those lenses.

III. Proposed Research

This dissertation employs a mixed-methods research design to address the three RQs. Accordingly, the proposed research comprises three main components: (1) benchmarking the trustworthiness of SOTA AI-based systems in SE, (2) trustworthiness testing via semantic analysis, and (3) improving the trustworthiness by utilising the test results.

A. ***RQ1****: Trustworthiness Benchmark*

We aim to construct a family of explanation-based trustworthiness benchmarks for AI-based systems in SE. Unlike existing benchmarks that record only output correctness, our benchmarks also capture each decision's explanation with its trustworthiness label. We collect decisions from SOTA systems across three SE tasks: NL text classification, vulnerability detection, and program repair. To ensure robustness, decisions are collected from both in- and out-of-distribution settings.

For each decision, we record the input, the system output, the output's correctness label, and explanations generated by various explainability methods. The next step is to label the decision's trustworthiness based on its explanation. Two experts are invited to independently do this. They are provided with the decision record and task-specific auxiliary information. They then review the explanation to determine whether the decision is trustworthy. Disagreements between the two experts are resolved by a third expert. The resulting benchmarks reveal to what extent the SOTA systems are trustworthy. We also perform qualitative analysis of the identified untrustworthy decisions to identify their characteristics and diagnose their root causes. The benchmarks help us to design effective approaches to test and improve the trustworthiness, as well as support systematic evaluation of these approaches.

B. ***RQ2****: Automated Trustworthiness Testing*

We conceptualise trustworthiness testing as the process of providing a system with inputs and observing its outputs and the corresponding explanations. To automate trustworthiness testing, one of the main challenges is the oracle problem. In particular, given a system's output and explanation, we need to automatically identify if this decision is trustworthy. Hence, to automate trustworthiness testing, we aim to address the trustworthiness oracle problem.

As the nature of explanations differs between classification and generation systems, we propose two respective directions.

1) Classification systems: The key idea is to analyse the semantic relationship between the elements that contribute the most to the decision and the output class. Such elements are identified by the explanations. For example, in NL text classification, explanations can highlight words. These words are then compared against the output class to determine whether the system relies on related words. In vulnerability detection, explanations can highlight suspicious lines of code. These lines can be assessed based on their syntactic meanings in comparison with historical patterns, as well as their code dependencies. A decision is considered untrustworthy when its explanation highlights irrelevant words or code regions that should not justify the decision.

2) Generative systems: These systems, such as LLMs, can provide CoT explanations expressed in NL. In this case, the proposed approach focuses on whether a CoT explanation correctly justifies the generated output. This can be broken down into semantic consistencies between the generated output and the explanation, between the explanation and the input context, and among the reasoning steps in the explanation. A decision is flagged as untrustworthy if any of these consistencies resurface. This allows the proposed approach to identify untrustworthy decisions without knowing the ground truths.

C. ***RQ3**: Automatically Improving Trustworthiness*

The proposed testing approaches in RQ2 provide valuable test results. In particular, they help engineers detect trustworthiness issues in a system's behavior. The detected issues can reveal why the system makes decisions for the wrong reasons. By utilising these issues as feedback, this dissertation aims to further improve the trustworthiness of AI-based systems in SE. For smaller systems, the test results can guide data augmentation, retraining, representation learning, or explanation regularisation [11]. By doing so, the system learns to reduce attention to spurious patterns and strengthen reliance on genuine task-relevant evidence. For larger systems, such as LLMs, the test results can support fine-tuning or model editing to correct unreliable reasoning.

Automated trustworthiness improving closes the loop between testing and repair. Automated trustworthiness testing identifies problematic reasoning, while improving techniques use this as feedback to make future decisions more trustworthy.

IV. Expected Contributions

The proposed research investigates the trustworthiness of AI-based systems in SE, as well as how to test and improve their trustworthiness through the following contributions:

- Manual-labeled datasets that benchmark the trustworthiness of SOTA AI-based systems. These datasets cover three SE tasks: NL text classification, vulnerability detection, and program repair. They can yield taxonomies and empirical insights into the prevalence and nature of trustworthiness. They also support the evaluation of trustworthiness testing, trustworthiness improving, and other trustworthiness-related research.
- Novel automated approaches for testing and improving the trustworthiness of AI-based systems in SE.
- Open-source tools that implement these approaches. These tools can be easily integrated into the development pipeline of AI-based systems.

Several studies from this dissertation have been accepted to the ACM International Conference on the Foundations of Software Engineering (FSE) 2025 [12] and IEEE Transactions on Software Engineering (IEEE TSE) [10]. We will continue to submit future work to top venues.

V. Results Achieved So Far

Our preliminary work benchmarks trustworthiness and explores the feasibility and effectiveness of the proposed approaches. We organize the results into three components corresponding to three RQs: trustworthiness benchmark, trustworthiness testing, and trustworthiness improving.

A. ***RQ1**: Trustworthiness Benchmark*

For classification tasks, we focus on benchmarking two tasks: NL text classification and vulnerability detection. We constructed *TokiBench* [12], a trustworthiness benchmark for NL text classifiers. TokiBench covers multiple domains, such as issue management and sentiment analysis. We also constructed *UntrustVulBench* [10] for vulnerability detection. These benchmarks contain around 123K decisions. These decisions are generated by multiple SOTA systems on multiple task-specific datasets. The benchmarks reveal that 13.5% of the decisions are actually untrustworthy. They also show that model confidence is not a reliable trustworthiness indicator. Qualitative analysis further shows that the main reason for untrustworthiness is the system's reliance on spurious correlations learnt during training.

For generation tasks, this dissertation focuses on program repair. We constructed *VFXplainBench*, a trustworthiness benchmark for LLM-based vulnerability repair. *VFXplainBench* contains 11,210 decisions generated by multiple SOTA systems, including both closed-source and open-source LLMs. VFXplainBench covers 1,121 vulnerabilities across 57 vulnerability types. Analysis from this benchmark shows that 9,821 decisions (87.6%) are untrustworthy. It also identifies three semantic inconsistencies that cause untrustworthiness. There are inconsistencies between the generated fixed program and its explanation, within the explanation's reasoning steps, and between the explanation and the vulnerable program.

B. ***RQ2**: Automated Trustworthiness Testing*

We developed *TOKI* [12], an automated approach for testing the trustworthiness of text classifiers. TOKI checks whether the words contributing the most to a decision are semantically related to the predicted class. Specifically, we leverage explainability methods to extract the decision-contributing words and measure their semantic relatedness with the class based on word embeddings. We evaluated TOKI's ability to differentiate trustworthy decisions from untrustworthy ones on TokiBench. Results show that TOKI achieves 142% higher accuracy than the confidence-based baseline.

We introduced *UntrustVul* [10], a new approach for identifying untrustworthy vulnerability predictions. We evaluated it on UntrustVulBench. Results show that UntrustVul effectively detects untrustworthy predictions with F1-score of 82%–94%.

We proposed *SusVF*, an automated method that detects untrustworthy decisions made by LLM-based program repair systems. It identifies semantic inconsistencies between the generated fixed program and its explanation, within the explanation's reasoning steps, and between the explanation and the vulnerable program. Evaluated on VFXplainBench, SusVF achieves 0.882 precision, 0.732 recall, substantially outperforming LLM-as-a-Judge baselines.

These results highlight the importance and effectiveness of explanation-based trustworthiness testing for safer deployment of AI-based systems in SE.

C. *RQ3: Automatically Improving Trustworthiness*

We developed an approach that improves the trustworthiness of classification systems. In particular, we applied causal learning to improve vulnerability detectors [10]. This aims to disable the detectors from using spurious features, and encourage the use of correct features. This approach improves the vanilla detectors in terms of both task performance by 321% and trustworthiness by 100%.

VI. Evaluation Plan

A. Evaluation of Benchmarks

The constructed datasets are manually labeled by human experts. To evaluate the labeling process, we report Cohen's kappa, a statistical metric used to measure inter-annotator agreement. We also report the proportion of trustworthy and untrustworthy decisions across systems and datasets.

B. Evaluation of Trustworthiness Testing

The proposed approaches for automated trustworthiness testing, such as TOKI, UntrustVul, and SusVF, differentiate untrustworthy decisions from trustworthy ones. To evaluate them, we formulate this task as a binary classification task, with two labels: trustworthy and untrustworthy. We then report standard binary classification metrics, such as accuracy, precision, recall, F1-score. Each approach is evaluated on the respective benchmark, which contains both in-distribution and out-of-distribution data to ensure robust evaluation. We compare each approach with baselines such as confidence-based or LLM-as-a-Judge. We also perform manual analysis of each approach's failure cases to identify its limitations.

C. Evaluation of Improving Trustworthiness

We will evaluate whether trustworthiness testing can guide the improvement of AI-based SE systems by using detected untrustworthy predictions as feedback. The improved systems will be compared with their vanilla versions and existing baselines on both in-distribution and out-of-distribution datasets. We will measure the changes in conventional task performance using precision, recall, and F1-score, and in trustworthiness using the constructed benchmarks. This evaluation will show whether the proposed approach can improve not only the general task-solving ability, but also its generalisability to out-of-distribution data and its trustworthiness.

VII. Plan For Demonstrating Research Contribution

This research aims to produce rigorous and practically valuable contributions to the SE community. To demonstrate its impact, the following strategies have been employed:

- The benchmarks, such as TokiBench, UntrustVulBench, and VFXplainBench, have been derived from decisions of SOTA systems on real-world data. We have also compared the proposed approaches and tools, such as TOKI, UntrustVul, and SusVF, against existing SOTA approaches. These comparisons highlight the novelty and practical advantages of the proposed research.
- We have released our datasets and tools as open-source projects to support reproducibility, collaboration, and broader adoption.
- Our future dissemination efforts will focus on engaging with academic venues and industrial communities. The work of TokiBench and TOKI has been accepted and was presented at FSE 2025 [12]. The work of UntrustVulBench and UntrustVul has been accepted at IEEE TSE [10], and will be submitted for presentation at a conference as a Journal-First paper. We will also continue to submit future work to leading venues.

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